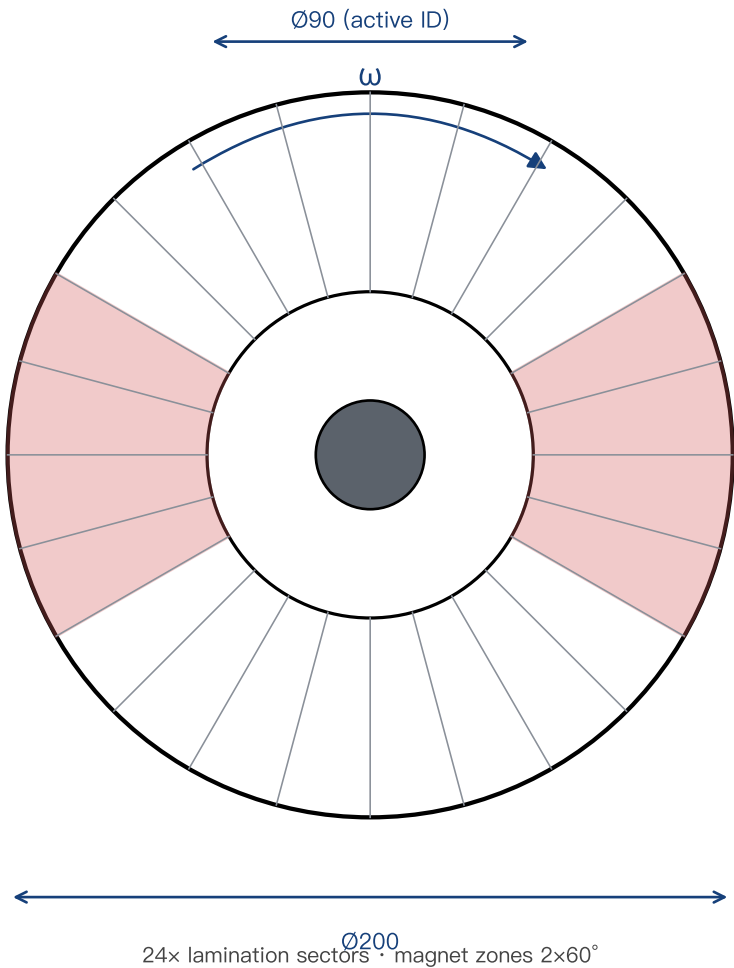
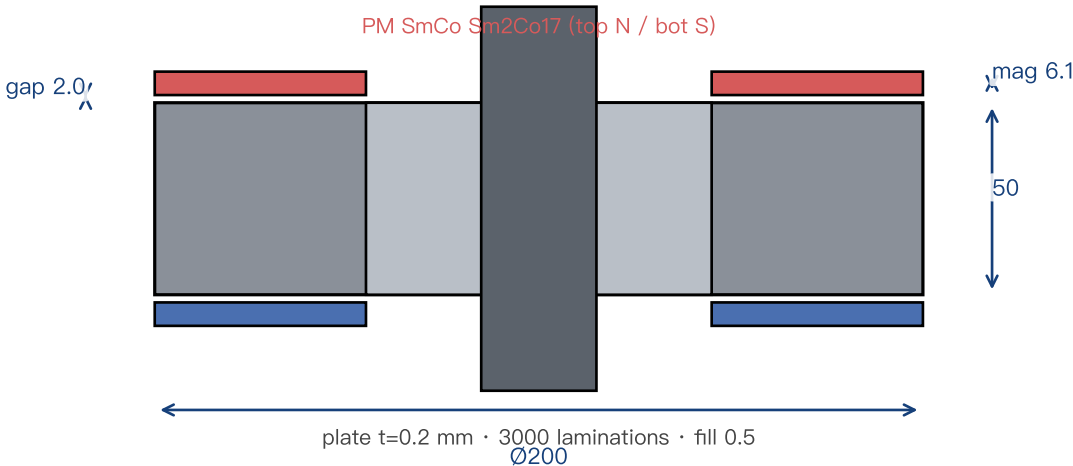


Flywheel Thermomagnetic Generator — Manufacturing Drawing (estimates; FEM/DFM review required)

PLAN VIEW (俯視)



SECTION A-A (剖面)



BILL OF MATERIALS / 成本估算

#	Part	Material	Qty	Mass kg	Cost \$
1	Active ring (laminations)	Fe–Co–Ni–Al 0.2mm	3000	4.7	141
2	Permanent magnets	SmCo Sm2Co17	4	1.24	149
3	Rotor hub + disc	Steel (M270)	1	4.2	13
4	Stator coil	Cu Ø0.8mm x400t	1	0.57	7
5	Bearings / seals	—	2	—	40
6	Fasteners / misc	—	—	—	30
	TOTAL			10.7	379

NOTES / GD&T:

- 疊片：0.2 mm 矽鋼/Fe–Co 軟磁箔，層間絕緣，疊壓係數 ≥0.95。
- 動平衡 G2.5 @ 1015 rpm；轉子同軸度 Ø0.02 A | 軸承位 Ø0.01。
- 氣隙 2.0±0.05 mm（含板）；磁體面平行度 0.02。
- 磁體 SmCo Sm2Co17，工作溫 ≤300°C（需熱隔離/冷側）；退磁裕度 50°C。
- 表面：軸承位 Ra0.8，其餘 Ra3.2。未注公差 ISO 2768–m。
- 回熱器（逆流，ε≈0.95）與熱/冷流道介面另見組裝圖。

PROJECT	alloy_engine Thermomagnetic Generator
PART	Flywheel TMG rotor assembly (Curie wheel)
DESIGN PT	350°C · Fe31Co23Ni17Al14
PERF	P≈3447W(ceiling) · η 0.477% · 1015rpm · 30.16Vrms
MASS / COST	10.7 kg · ~\$379
UNITS / SCALE	mm · NTS